

# RC Design Project 01

We can delete this since it no need



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## 1 How to Design

- Building Load contains : **gravity** and **lateral** Loads.



## 2 Introduction

This is Our Team Intro.

### 3 Detailed Design Calculations

#### 3.1 Loading

##### 3.1.1 Live Load

For imposed loads on floor according to Table 6.1 – 6.4 in EN 1991-1-1:2002 [3]. Is summary in Table 3.1.

Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

| Category          | Sub-category / Use                                                                  | $q_k$<br>[kN/m <sup>2</sup> ] | $Q_k$<br>[kN]       |
|-------------------|-------------------------------------------------------------------------------------|-------------------------------|---------------------|
| <b>Category A</b> | Areas for domestic and residential activities                                       |                               |                     |
|                   | - Floors                                                                            | 1,5 to 2,0                    | 2,0 to 3,0          |
|                   | - Stairs                                                                            | 2,0 to 4,0                    | 2,0 to 4,0          |
|                   | - Balconies                                                                         | 2,5 to 4,0                    | 2,0 to 3,0          |
| <b>Category B</b> | Office areas                                                                        | 2,0 to 3,0                    | 1,5 to 4,5          |
| <b>Category C</b> | Areas where people may congregate                                                   |                               |                     |
| C1                | Areas with tables<br>(e.g. schools, cafes, restaurants, reading rooms)              | 2,0 to 3,0                    | 3,0 to 4,0          |
| C2                | Areas with fixed seats<br>(e.g. theatres, cinemas, churches)                        | 3,0 to 4,0                    | 2,5 to 7,0<br>(4,0) |
| C3                | Areas without obstacles for moving people<br>(e.g. museums, entrance halls)         | 3,0 to 5,0                    | 4,0 to 7,0          |
| C4                | Areas with possible physical activities<br>(e.g. dance halls, gyms)                 | 3,5 to 5,0                    | 3,5 to 7,0          |
| C5                | Areas susceptible to large crowds<br>(e.g. concert halls, sports halls with stands) | 5,0 to 7,5                    | 3,5 to 4,5          |
| <b>Category D</b> | Shopping areas                                                                      |                               |                     |

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Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

| Category          | Sub-category / Use                                                       | $q_k$<br>[kN/m <sup>2</sup> ] | $Q_k$<br>[kN]       |
|-------------------|--------------------------------------------------------------------------|-------------------------------|---------------------|
| D1                | Areas in general retail shops                                            | 4,0 to 5,0                    | 3,5 to 7,0<br>(4,0) |
| D2                | Areas in department stores                                               | 4,0 to 5,0                    | 3,5 to 7,0          |
| <b>Category E</b> | Storage and industrial activities<br>(general storage, accessible areas) | 7,5                           | 7,0                 |

And this is the imposed loads on second floor that we assumed. which shown in Table 3.2.

Table 3.2: Imposed Loads On Second Floor

| No. | Room / Area                      | Category<br>(EN 1991-1-1) | $q_k$<br>(kN/m <sup>2</sup> ) |
|-----|----------------------------------|---------------------------|-------------------------------|
| 1   | Receptionist/<br>Reception       | B                         | 3                             |
| 2   | Workstation /<br>Open office     | B                         | 3                             |
| 3   | Meeting Room                     | C1                        | 3                             |
| 4   | President Office                 | B                         | 3                             |
| 5   | Vice President Offices           | B                         | 3                             |
| 6   | Accounting                       | B                         | 3                             |
| 7   | General Sale &<br>Administration | B                         | 3                             |
| 8   | Copy /<br>Printer Area           | B                         | 3                             |
| 9   | Storages                         | E1                        | 5                             |
| 10  | Lift Lobby /<br>Lift Area        | C3                        | 4                             |
| 11  | Stair (Main Stair)               | A                         | 4                             |
| 12  | Restroom (Men)                   | B                         | 3                             |
| 13  | Restroom (Women)                 | B                         | 3                             |
| 14  | —                                | —                         | —                             |
| 15  | —                                | —                         | —                             |
| 16  | —                                | —                         | —                             |

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Table 3.2: Imposed Loads On Second Floor

| No. | Room / Area                | Category<br>(EN 1991-1-1) | $q_k$<br>(kN/m <sup>2</sup> ) |
|-----|----------------------------|---------------------------|-------------------------------|
| 17  | Entrance                   | C3                        | 4                             |
| 18  | Emergency Stair            | A                         | 4                             |
| 19  | Server &<br>Technical Room | E1<br>(conservative)      | 5                             |
| 20  | Kitchen /<br>Pantry        | C1                        | 3                             |
| 21  | Library                    | C3                        | 4                             |
| 22  | Share Space                | C3                        | 4                             |

### 3.1.2 Dead Load

| Source                                 | Typical Value |
|----------------------------------------|---------------|
| Reinforced concrete slab (200mm thick) |               |
| Screed/floor finish                    |               |
| Ceiling + services (MEP)               |               |
| Partitions (treated as permanent)      |               |
| <b>Total G<sub>k</sub></b>             |               |

## References

- [1] BSI, *BS EN 1992-1-1:2023 - Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings, bridges and civil engineering structures*, BSI Standards Publication, 2023.
- [2] BSI, *BS EN 1991-1-1:2025 - Eurocode 1: Actions on structures - Part 1-1: Specific weight of materials, self-weight of construction works and imposed loads for buildings*, BSI Standards Publication, 2025.
- [3] British Standards Institution, *BS EN 1991-1-1:2002 - Eurocode 1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings*, Incorporating corrigenda December 2004 and March 2009, BSI, 2002.
- [4] CEN, *EN 1991-1-4:2005+A1 - Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, European Standard, Incorporating corrigendum January 2010, April 2010.
- [5] CEN, *Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, EN 1991-1-4:2005+A1, English version, April 2010.

*Note that :* We may use the referennnces provide by professor (check in telegram group) if it still not enough we can add more maybe.

# Appendices

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then try